

Vikram Anantha

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EDUCATION

University of Michigan — B.S. Computer Science, 1931E Scholar

Aug. 2024 - May 2027 (expected)

- **Relevant course work:** Machine Learning, Computer Security, Data Structures & Algorithms, Digital Circuits, Computer Organization, Linear Algebra, Discrete Math

3.96 / 4.00 GPA

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JS, HTML / CSS, CAPL, React, Assembly, Bash, Markdown, LaTeX

Frameworks & Tools: Robot Operating System (ROS), Amazon Web Services (AWS), Vector CANalyzer, MATLAB/Simulink, OpenCV, TensorFlow/PyTorch, Git/GitHub, CUDA, Docker

AI/ML Algorithms: Gaussian Splatting, LLMs, YOLO v8 Object Detection, SAM-2 Segmentation, SLAM

WORK EXPERIENCE

Capoom — Software Eng. Intern

Jan. 2025 - Present

- Architected an automated pipeline for 3D Gaussian Splat model creation of UofM's Mcity
- Implemented hierarchical methods to visualize Gaussian Splat map

Stellantis — ADAS Functional Integration Eng. Intern

May 2025 - Aug. 2025

- Deployed a CAPL script to diagnose CAN-bus anomalies, identifying a 10% vehicle-component fault rate
- Designed and validated MATLAB Simulink model to control CAN-bus signals of vehicle to quantify the self-driving feel

Constant Therapy Health — Software Eng. Intern

Jun. 2024 - Aug. 2024

- Developed Automatic Speech Recognition model for stroke aphasia patients & web-based hierarchical task visualizer
- Reduced false negative rate by 4x compared to Google's Speech Recognition Engine

RESEARCH EXPERIENCE & PROJECTS

University of Michigan, ROAHM Robotics Lab

Aug. 2024 - Present

- Creating a real-time 3D object segmented model of a changing env. in Gaussian Splat for robotic arm movement
- Developing new Transformer-based Vision Language Action models to create faster generalist robot policies

Columbia University, Center for Smart Street Scapes

May 2023 - Feb. 2024

- Worked on NSF Project, developed app to warn pedestrians at road intersections of oncoming vehicles
- Reduced latency to <2s for human reaction time to prevent accidents using React Native and MQTT messaging

MIT, Auto ID Lab

Jun. 2023 - Sep. 2023

- Worked on 2FA system to prevent hacking for autonomous vehicles at road intersections
- Used the YOLOv8 Computer Vision model and Python to recognize vehicles and headlight patterns

HELM Learning — Co-founder, Teacher, Developer

Apr. 2020 - May 2024

- Created free online peer-to-peer worldwide learning platform with over 4.2K students, 90+ classes
- Created services with Python, JS, SQL, Recommendation ML Algorithm, hosted platform on AWS

SELECTED HONORS & AWARDS

1931e Merit Scholarship, University of Michigan - 1 of 6 recipients

May 2025

Diamond Challenge, Regionals - 1st Place Prize Award

Mar. 2023

American Invitational Mathematics Exam (AIME) - 2 Time Qualifier

Mar. 2021 & Feb. 2022

USA Coding Olympiad (USACO) - Silver Division

Dec. 2019

PUBLICATIONS

V. Anantha, et al., "Digital Twin for Pedestrian Safety Warning at a Single Urban Traffic Intersection," 2024 IEEE Intelligent Vehicles Symposium (IV), pp. 2640-2645, [doi: 10.1109/IV55156.2024.10588544](https://doi.org/10.1109/IV55156.2024.10588544)

V. Anantha, Peer to Peer Learning Platform Optimized With Machine Learning, [arXiv:2209.03489](https://arxiv.org/abs/2209.03489), 2022